DRUVA TEJA C

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PROFESSIONAL SUMMARY

Highly motivated first-year Computer Science & Engineering B. Tech undergraduate. Skilled in Python, SQL, FastAPI, and JavaScript, with hands-on experience building full-stack projects, including a REST backend with a MySQL database and an AI computer-vision pipeline (OpenCV, YOLOv8). Strong interest in software architecture, cybersecurity, and clean coding practices. Seeking an SDE Internship or Junior Developer opportunity to apply problem-solving ability and a builder's mindset to real-world software projects.

EDUCATION

NxtWave Institute of Advanced Technologies (NIAT), S-VYASA University *Aug 2026 – Present (Expected 2030)*
Bachelor of Technology (B.Tech) — Computer Science & Engineering

- 1st-Year Student

Sri Kumaran Children's Home Pre-University College *Jun 2024 – Mar 2026*
Pre-University Course (PCMCs: Physics, Chemistry, Mathematics, Computer Science)

Holy Saint High School
Secondary School Leaving Certificate (10th Standard)

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
Web & Database: HTML, CSS, FastAPI, Node.js/Express, REST APIs, MySQL, SQLAlchemy, JWT authentication, responsive web design
AI & Geospatial: OpenCV, YOLOv8 (Ultralytics), Google Maps JavaScript API, geospatial analysis (Haversine, point-in-polygon)
Tools & Platforms: VS Code, Git/Github basics, Adobe Photoshop, Adobe Premiere Pro
Core Interests: Software Architecture, Cybersecurity, Clean Code Practices, Data Structures & Algorithms, Problem Solving

ACADEMIC & PERSONAL PROJECTS

Multi-Page Preschool Website — Full-Stack Development *2026*
Independent Project | HTML, CSS, JavaScript, Node.js, Express, MySQL

- Built and deployed a five-page responsive website (Home, About, Academics, Gallery, Contact) from initial HTML structure through a full responsive-design overhaul across mobile, tablet, and desktop breakpoints.
- Developed a Node.js and Express backend with a MySQL database, implementing a lead-capture API endpoint and a Basic Auth-protected admin route for managing form submissions.
- Engineered a reusable shared-header component using a JavaScript fetch-based include pattern, reducing duplicated markup across pages and simplifying future maintenance.

MobiSense CityVision — AI-Powered Municipal Incident Detection Platform *2026*
Team Project, Smart India Hackathon 2026 | Backend & Database Developer | Python, FastAPI, MySQL, YOLOv8, JWT, JavaScript

- Designed the MySQL 8.0 database (issue types, verified employees, users, issues, raw detections, status history) and stored procedures, including a spatial de-duplication engine using ST_Distance_Sphere with per-category radii and a running centroid, so repeated detections of one pothole, garbage spot, or accident merge into a single issue.
- Built a FastAPI REST backend with bcrypt password hashing and JWT authentication; signup is restricted to whitelisted government employee IDs enforced through a foreign key. Endpoints cover issues, status updates, detection ingestion, SLA analytics, and recurring hotspots.
- Added SQL views for SLA tracking (hours open, overdue flag, resolution time per category) and recurring-hotspot detection, plus an accident alert log that routes alerts to the relevant departments.
- Integrated YOLOv8 detection scripts (potholes, garbage, traffic density, and a bounding-box-overlap heuristic for accidents) with an ingest client that buffers detections offline and replays them in chronological order when connectivity returns; connected the JavaScript dashboard (live city map, incident center, fleet, analytics) to the API.

MobiSense Watershed Intelligence — Geospatial Decision-Support Platform *2026*
Prototype for SIH 2026 PS SIH26015 | Python, FastAPI, SQLAlchemy, MySQL/SQLite, OpenCV, YOLO, Google Maps API, JavaScript

- Built a prototype that turns geo-tagged field photographs into watershed insights: extracts GPS coordinates from EXIF metadata, maps each photo to its watershed boundary, and runs AI feature detection on the image.
- Developed a FastAPI backend with 20+ REST endpoints using SQLAlchemy 2.0 ORM and Pydantic v2 validation; runs on MySQL 8.0 with an automatic fallback to SQLite when MySQL is unavailable.
- Implemented spatial and temporal analytics: Haversine distance, ray-casting point-in-polygon containment, and before/after change detection on vegetation and water indicators.

- Created an OpenCV pipeline (Laplacian sharpness check, CLAHE contrast normalisation, HSV segmentation of vegetation, water, and soil, contour-based bounding boxes) behind a modular YOLO detector interface.
- Delivered an 8-page vanilla JavaScript dashboard with Google Maps polygons, canvas charts, a before/after comparison slider, and printable reports, backed by automated API, upload, and frontend test scripts.

HACKATHONS & LEADERSHIP

Advanced Tech Club — NIAT, S-VYASA Bangalore *Oct 2026 – Present*

Vice President

- Appointed as Vice President of the Advanced Tech Club, a student-run technical club at NIAT, S-VYASA Bangalore, responsible for planning and executing club events and activities.
- Lead a core team of five department heads (Operations, Outreach, Creative & Design, Finance, and Social Media), delegating responsibilities and coordinating their work across event planning and execution.

Smart India Hackathon (SIH) 2026 — College-Level Internal Hackathon *Sep 2026*

Team Lead | Problem Statement SIH26124 — Bharat Electronics Limited (BEL), Smart Automation

- Led the team as Team Lead through ideation, solution design, and presentation for PS SIH26124 (BEL) under the Smart Automation theme at the college-internal round of Smart India Hackathon 2026.
- Selected to represent the college in the SIH screening round; prepared and submitted the team's solution presentation (PPT) on the official SIH portal as Team Lead.

Smart India Hackathon (SIH) 2026 — College-Level Internal Hackathon *Sep 2026*

Team Lead | Problem Statement SIH26015 — Ministry of Rural Development (MoRD), Geo-Coded Images for Watershed Development

- Led the team as Team Lead on PS SIH26015, which asks for geospatial techniques to visualise and analyse geo-coded images and improve watershed development outcomes.
- Directed the build of the MobiSense Watershed Intelligence prototype for this problem statement: EXIF GPS extraction, watershed boundary mapping, OpenCV/YOLO feature detection, and before/after change detection on a Google Maps dashboard.

InnovateX — Intra-College Hackathon *Sep 2026*

Team Lead | Problem Statement SIH26124 — Bharat Electronics Limited (BEL), Smart Automation

- Led the team as Team Lead in the college's intra-college hackathon on the same problem statement (SIH26124, Smart Automation), guiding solution iteration and final presentation in a fast-paced, time-boxed setting.

EXTRACURRICULAR ACTIVITIES & CERTIFICATIONS

- Actively developing skills in JavaScript and backend web development in preparation for the B.Tech coursework.
- Self-directed learner: built and shipped working web projects independently ahead of formal coursework, demonstrating initiative and practical application of classroom-adjacent skills.